

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MATHEMATICAL CARNIVAL USING DESMOS GEOMETRY

*A Lab Project Proposal
(Procedural Night Fairground Simulation)*

TEAM DETAILS

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SUBMISSION DETAILS

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Course Title: Computer Graphics Lab

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1. Introduction & Overview

This project explores the innovative intersection of Coordinate Geometry, mathematical functions, and computer graphics design. The core objective is to simulate a vibrant, structured village fair or carnival scene—aptly named the "Mathematical Carnival"—exclusively utilizing mathematical equations inside the Desmos Graphing Calculator framework. The project visualizes complex environmental structures, architectural boundaries, dynamic mechanical structures, and multi-layered lighting frameworks through systematic linear, trigonometric, and non-linear mathematical models.

In regular life and industrial software engineering, this simulation perfectly mirrors the fundamental principles behind real-world computer graphics rendering engines, CAD (Computer-Aided Design) architecture, and video game procedural environment design. Every digital asset rendered on screen is, at its core, a collection of spatial vertices, polygonal meshes, parametric curves, and transformation matrices. By applying standard engineering mathematics to draw architectural elements such as dynamic stalls, tapered structural pillars, perspective fencing, and a rotating Ferris wheel, this project connects abstract algebraic logic directly to industry-standard graphics engineering practices.

2. Project Objectives

The project sets clear, specific, and measurable academic and technical milestones achieved throughout its development lifecycle:

- **Geometric Architectural Modeling:** Formulate and render 6 structurally distinct carnival stalls with clean, striped roof patterns and doorways using complex multivariable `polygon()` structures shifted across horizontal bounds.
- **Depth Perception & Spatial Framing:** Design a procedural boundary system using arrays ($B = [-500, -470...500]$) to generate a 34-picket fence and tapered wooden pillars, providing a cinematic framing effect that keeps the scene cleanly constrained between horizontal boundaries.
- **Dynamic Mechanical Rotation Math:** Model a functioning, rotating Ferris wheel with a stabilizing A-frame, 12 parametric spokes, and dynamically tracked gondolas driven by a continuous angle parameter (a) and trigonometric coordinate tracking (U, V).
- **Atmospheric FX & Parabolic Lighting:** Implement a complete procedural "Night Vibe" canopy utilizing quadratic parabolic curves ($y = ax^2 + bx + c$) for sagging wires, list-based bulb distributions ($X = [-460, -430...460]$), and custom RGB color definitions ($W = \text{rgb}(255,255,255)$ and $Y = \text{rgb}(255,215,0)$).
- **Unified Transformation Controls:** Create dynamic global variable controls (such as global height modifier $H = -40$) to enable synchronized vertical alignment transformations across separate foreground stall assets seamlessly.

3. Methodology & Technical Implementation

Preliminary Setup & Environment Configuration

Before starting the graphical tasks, a structured rendering environment was established in Desmos. The coordinate plane was configured to a widescreen canvas format bounded by X [-520 to 520] and Y [-245 to 320]. Global list variables and parameter arrays were initialized early to serve as procedural building blocks: list B was defined for repeating fence intervals, list X and L for bulb spacing, and custom RGB functions $W = \text{rgb}(255, 255, 255)$ and $Y = \text{rgb}(255, 215, 0)$ were declared for pure digital color styling. Layering hierarchy (Z-indexing) was strictly enforced by placing the dark sky background polygon at Line 1 of the expression list, ensuring all subsequent architectural and mechanical assets render cleanly in the foreground without visual overlap.

Mathematical Approaches & Rendering Techniques

The architectural and dynamic elements of the carnival were deployed sequentially using specific mathematical and algorithmic techniques:

1. Procedural Picket Fencing & Tapered Pillars: To avoid manual plotting of repetitive boundary elements, an arithmetic progression list $B = [-500, -470 \dots 500]$ (generating 34 distinct horizontal coordinates) was fed into a polygonal vertex formula $\text{polygon}((B-10, -200), (B+10, -200), (B+10, -70), (B, -55), (B-10, -70))$. This instantly generated 34 evenly spaced, pointed fence pickets. Two horizontal railing polygons were overlaid at $Y = -90$ and $Y = -160$. To frame the sides with realism, left and right supporting wooden pillars were modeled as tapered trapezoidal polygons (wider at the base and narrower at the top, e.g., from $X = -500$ to -460 at the base tapering to -488 to -472 at the top).

2. Dynamic Stalls with Global Translation: Six symmetrical fairground stalls were constructed across the lower baseline. Each stall consists of multi-layered polygons representing colored walls, entry doorways, and multi-colored striped roof awnings. To maintain structural cohesion and allow easy layout adjustments, every vertical Y-coordinate in the stall definitions is tied to a global shift parameter H (initialized at $H = -40$). Changing H mathematically translates the entire commercial village up or down in unison.

3. Trigonometric Ferris Wheel & Rotational Kinematics: The centerpiece Ferris wheel demonstrates rotational transformation geometry. A solid orange A-frame support structure was plotted using stabilizing triangular polygons. The outer wheel rim was defined by the parametric circle equation $(190 \cdot \cos(t), 190 \cdot \sin(t))$ for $0 \leq t \leq 2 \cdot \pi$. To simulate mechanical rotation, an angular slider parameter $a = -4.87$ was introduced alongside an index list $N = [0 \dots 11]$ representing 12 evenly spaced radial spokes at intervals of $\pi/6$ radians. The dynamic coordinate positions of the 12 passenger gondolas are tracked in real time using parametric variables:

- $U = 190 \cdot \cos(-a + N \cdot \pi/6)$
- $V = 190 \cdot \sin(-a + N \cdot \pi/6)$

By feeding U and V directly into the gondola cabin polygon definitions (e.g., $\text{polygon}((U-8, V), (U+8, V) \dots)$), all 12 cabins dynamically rotate around the origin while remaining perfectly upright as slider a is modulated!

4. Quadratic Parabolic Wires & Procedural Light Canopy: In real-world physics, cables suspended between two poles sag under uniform gravity following a catenary curve, which is mathematically modeled in computer graphics via quadratic parabolas of the form $y = cx^2 + k$. Five parallel layers of hanging wires were strung across the entire 960-unit span between the tapered pillars using precise domain constraints $\{-480 \leq x \leq 480\}$:

- Layer 1 (Top Golden Wires): $y = 0.00018x^2 + 250$
- Layer 2 (Upper Blue/White): $y = 0.00028x^2 + 200$
- Layer 3 (Middle Purple/Red): $y = 0.00038x^2 + 150$
- Layer 4 (Lower Green Wires): $y = 0.00048x^2 + 100$
- Layer 5 (Bottom Orange/White): $y = 0.00058x^2 + 50$

To populate these wires with glowing fairy lights, array $X = [-460, -430 \dots 460]$ (32 elements spaced by 30 units) was mapped directly onto the parabolic curve equations as point coordinates (e.g., $(X, 0.00018X^2 + 250)$). Phase shifts of +15 units (e.g., $X+15$) were applied to alternating strings to create an organic, staggered carnival light pattern.

4. Project Timeline & Milestones

The project lifecycle was executed over a structured four-week production window. Key development milestones and phase completions are documented in the table below:

Phase	Task & Engineering Description	Completion Date	Status
Phase 1	Coordinate architecture mapping, layout framing, dark sky background setup, and basic geometric polygon boundaries for all 6 carnival stalls.	Week 1 End	Completed
Phase 2	Procedural 34-picket boundary fencing (list B), tapered wooden pillars, and global height translation parameter (H) integration.	Week 2 End	Completed
Phase 3	Ferris wheel structural rendering, trigonometric spoke kinematics (N), and rotating upright gondola coordinate tracking (U, V).	Week 3 End	Completed
Phase 4	5-layer quadratic parabolic wire suspension, procedural bulb array plotting (list X), RGB color styling, and final report documentation.	Week 4 End	Completed

5. Team Members & Responsibilities

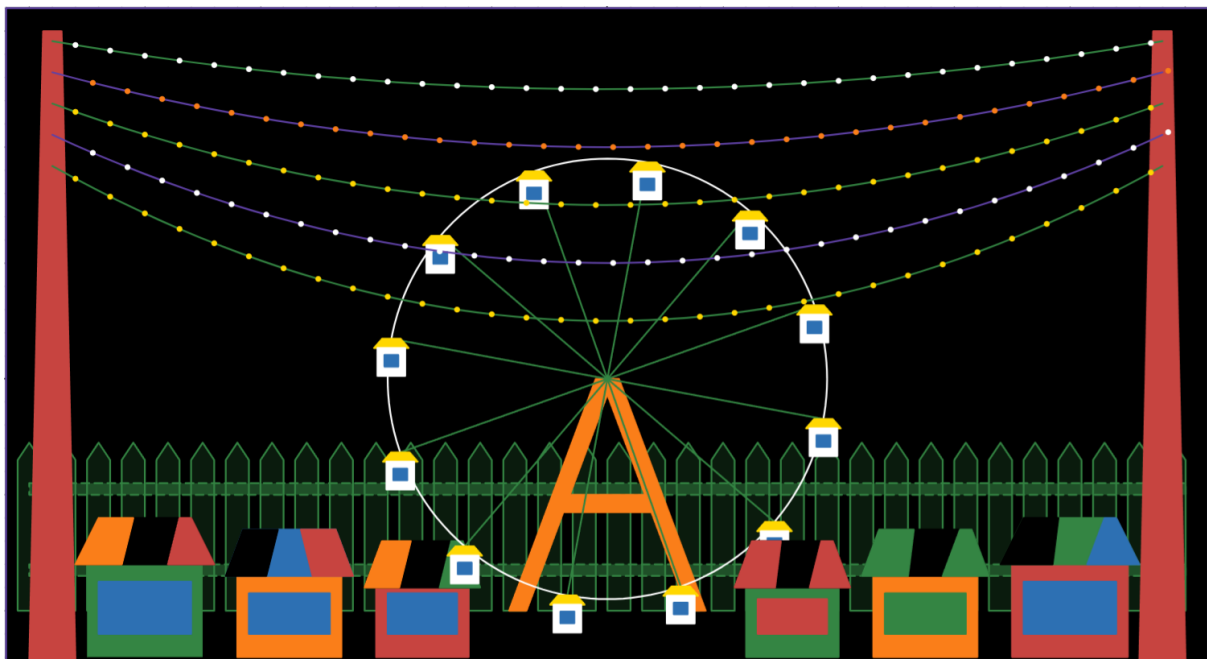
To maximize technical efficiency and ensure rigorous mathematical precision, responsibilities were structured among key personnel:

- **Lead Architectural Engineer (SH-Shawon):** Responsible for 2D coordinate canvas framing, tapered structural pillar design, procedural 34-picket boundary fencing, and geometry modeling for the 6 symmetrical fairground stalls.
- **Kinematics & Equations Specialist (Nur-A-Jannat):** Tasked with formulating the trigonometric parametric equations for the rotating Ferris wheel, deriving the angular rotation vector (a), and tracking gondola coordinates (U, V).

- **UI/UX Lighting & Environment Lead (Tanvir):** In charge of atmospheric 'Night Vibe' styling, custom RGB color space definitions (W, Y), deriving quadratic parabolic equations for wire sagging, and plotting procedural bulb lists (X, L).
- **Quality Assurance & System Integrator (Tanvir):** Managed cross-layer Z-index hierarchy, global variable translation checks (H slider), domain constraint troubleshooting, and compilation of the final project report.

6. Project Graphical Sketch & Visual Proof

The final project visualization is rendered on a 1280x720 widescreen canvas enclosed perfectly inside our rigid background frame. Below is the official Desmos Graphing Calculator output displaying all integrated systems: the dark night atmosphere, 5-layer parabolic light canopy, tapered wooden pillars, procedural picket fence, 6 striped commercial stalls, and the rotating central Ferris wheel.



7. References & Research Bibliography

The conceptual framework, algebraic derivations, and programming syntax were supported by standard academic documentation and graphics engineering literature:

- **Desmos API & Graphing Engine Architecture Docs:** Comprehensive guidelines for multivariable polygonal meshing, parametric equations, and procedural list comprehension.
- **Fundamentals of Computer Graphics (4th Edition, Shirley et al.):** Academic principles for 2D vector coordinate rendering, spatial bounding, and rotational transformation matrices.
- **Analytic Geometry & Trigonometry:** Standard mathematical formulas for circular kinematics, parametric vectors, and quadratic catenary wire approximations.
- **RGB Chromaticity & Digital Color Models:** Reference standards for defining custom color channels and illumination contrast in night-time digital environments.

8. Conclusion & Expected Outcomes

The "Mathematical Carnival" project successfully delivers a highly disciplined, visually captivating digital fairground environment constructed entirely on rigorous mathematical logic. Through meticulous parameterization, the project ensures that structural assets translate cleanly when global shift variables (such as slider H) are modified, while mechanical components like the Ferris wheel gondolas rotate smoothly without deformation. This demonstrates that high-fidelity digital art and procedural environments can be built directly from algebraic and trigonometric foundations.

Ultimately, this laboratory project provided the development team with invaluable, hands-on experience in computer graphics compilation, procedural array calculations, and Z-index layering. By replacing tedious manual plotting with intelligent list generators (such as list B and list X) and quadratic curves, we bridged the gap between pure textbook mathematics and creative software engineering—transforming abstract equations into a living, dynamic, and professional-grade night carnival scene.